Conditioning Thresholds for Lyapunov Metrics on Near-Peripheral Jordan Families

Why Global Weighting Must Follow Directional Deflation

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Abstract

RH-64 replaced Euclidean propagation of a terminal Arnoldi residual by the canonical Lyapunov metric $M - A^*MA = I$. Finite-dimensional contraction is automatic, but its family-level cost was left open. This paper identifies the relevant threshold on the model family

$$A_s = \sqrt{1-s}\,I + c_s N_d,$$

where N_d is the nilpotent upper shift and $s \downarrow 0$ is the peripheral gap. First, for every stable matrix,

$$q_M^2 = \left\| M^{1/2} A M^{-1/2} \right\|^2 = 1 - \lambda_{\max}(M)^{-1}.$$

Thus metric conditioning and metric contraction are two entries of one Gramian ledger. For fixed d, if $c_s = O(s)$, then $M_s \asymp s^{-1} I$, $\operatorname{cond}(M_s) = O(1)$, and $1 - q_{M,s} \asymp s$. If instead $c_s = \kappa s^\alpha$ with $0 \le \alpha < 1$, then

$$\operatorname{cond}(M_s) \gtrsim s^{-2(d-1)(1-\alpha)}, \qquad 1 - q_{M,s} \lesssim s^{1+2(d-1)(1-\alpha)}.$$

For fixed coupling, a logarithmically growing chain already gives super-polynomial conditioning.

A 140-decimal-digit audit recovers all predicted powers: in dimension four, fixed coupling gives fitted exponents 6.000004 and 7.000002, while gap-scale coupling gives bounded conditioning and a first-power contraction gap. A 256-bit Arb calculation certifies the exact two-step witness. The conclusion is a localized no-go result: a full-space Lyapunov metric cannot replace slow-direction removal, and even the matched regime retains an s^{-1} generic horizon. RH-64 remains viable after Krylov or block deflation has reduced the residual coupling and removed peripheral modes. No production physical-family uniformity, Stage A1 closure, arithmetic trace formula, or Hilbert–Polya conclusion is claimed.

Keywords: Lyapunov metric; Jordan chain; nonnormality; Gramian conditioning; Krylov residual; peripheral spectrum.

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1 Introduction

The finite-dimensional theorem of RH-64 is deliberately easy [3]: every stable matrix admits a positive Lyapunov metric in which it is a strict contraction [4]. The difficult question is whether those metrics can be compared uniformly as a physical small-noise family approaches its peripheral spectrum. Nonnormal chains are the minimal test because their spectral radius and transient amplification separate [2].

This paper audits the canonical right-hand side I. That choice is not claimed optimal, but it is exactly the metric used by RH-64 and exposes the cost of global weighting. The result is neither an abandonment of weighted tails nor a family theorem for the folded-Gaussian operators. It determines the order in which the existing tools must be applied:

$$\text{directional or block deflation} \longrightarrow \text{localized residual} \longrightarrow \text{weighted terminal propagation.} \tag{1}$$

Contributions and boundary

1. We prove an exact identity linking the weighted contraction to the largest eigenvalue of the Lyapunov metric.
2. We prove uniform conditioning when the Jordan coupling is at most the peripheral-gap scale.
3. We prove explicit conditioning, contraction-gap, and horizon obstructions above that scale, including a growing-dimension corollary.
4. We validate the power ledger at high precision and certify the two-step formulas with interval arithmetic.

The theorems concern finite Jordan families. The numerical rows are high-precision deterministic evidence, not a computation on the production physical family.

2 One metric ledger

Let A be a stable matrix and

$$M - A^*MA = I, \qquad M = \sum_{j\geq 0}(A^*)^j A^j. \tag{2}$$

Write $\|x\|_M = \left\|M^{1/2}x\right\|_2$ and $q_M = \left\|M^{1/2}AM^{-1/2}\right\|_2$.

Proposition 2.1 (Exact contraction ledger)**.** *The metric is positive, $M \succeq I$, and*

$$q_M^2 = 1 - \frac{1}{\lambda_{\max}(M)}, \qquad 1 - q_M = \frac{1}{\lambda_{\max}(M)(1+q_M)}. \tag{3}$$

Moreover transfer between Euclidean and metric norms costs at most $\sqrt{\mathrm{cond}(M)}$.

Proof. The series in (2) gives $M \succeq I$. Conjugating the Lyapunov identity by $M^{-1/2}$ gives

$$M^{-1/2}A^*MAM^{-1/2} = I - M^{-1}.$$

The largest eigenvalue on the right is $1 - \lambda_{\max}(M)^{-1}$, proving the first identity; rationalizing gives the second. The transfer statement is the extremal-eigenvalue comparison for a positive matrix [1]. □

The proposition shows why existence alone is insufficient. A large $\lambda_{\max}(M)$ simultaneously weakens $1 - q_M$ and can enlarge norm transfer. These are not separate losses to be optimized independently.

3 The coupling–gap threshold

Let N_d have ones on the first superdiagonal and zeros elsewhere. For $0 < s < 1$, define

$$A_s = q_s I + c_s N_d, \qquad q_s = \sqrt{1-s}, \qquad M_s - A_s^* M_s A_s = I. \tag{4}$$

The eigenvalues of A_s all equal q_s, independently of c_s.

Theorem 3.1 (Matched-scale conditioning). *Fix d and $\Gamma < \infty$. There are constants $C_-, C_+ > 0$ and $s_0 > 0$, depending only on (d, Γ), such that whenever $0 < s < s_0$ and $0 \le c_s \le \Gamma s$,*

$$C_- s^{-1} I \preceq M_s \preceq C_+ s^{-1} I. \tag{5}$$

Consequently

$$\operatorname{cond}(M_s) = O_{d,\Gamma}(1), \qquad 1 - q_{M,s} \asymp_{d,\Gamma} s. \tag{6}$$

Proof. Put $B_s = (c_s/q_s)N_d$. The finite binomial expansion gives

$$A_s^j = q_s^j \sum_{k=0}^{d-1} \binom{j}{k} B_s^k.$$

For fixed d, $c_s \le \Gamma s$, and $q_s \ge 1/2$, the squared norm is bounded by q_s^{2j} times a fixed polynomial in sj. The elementary moment estimates

$$\sum_{j \ge 0} (1-s)^j (sj)^m \le C_m s^{-1}$$

therefore imply the upper inequality in (5).

For $0 \le j \le s^{-1}$, the negative-binomial formula for $(I + B_s)^{-j}$ is a sum of only d terms, each uniformly bounded because $j \|B_s\| = O(1)$. Hence $\left\|(I + B_s)^j x\right\| \ge C^{-1} \|x\|$ on this range. Also $q_s^{2j} = (1-s)^j$ is bounded below there. Summing these first $\lfloor s^{-1} \rfloor + 1$ terms of $x^* M_s x = \sum_j \left\|A_s^j x\right\|^2$ proves the lower inequality. The conclusions follow from Proposition 2.1 and $\lambda_{\max}(M_s) \asymp s^{-1}$. □

The next estimate makes the opposite regime explicit.

Theorem 3.2 (Unmatched Jordan obstruction). *Let $d \ge 2$, $c_s > 0$, and*

$$0 < s \le \min\left\{\frac{1}{2}, \frac{1}{2(d-1)}\right\}.$$

Then

$$\lambda_{\max}(M_s) \ge \frac{c_s^{2d-2}}{2^{2d+4}((d-1)!)^2 s^{2d-1}}, \tag{7}$$

$$\operatorname{cond}(M_s) \ge \frac{c_s^{2d-2}}{2^{2d+4}((d-1)!)^2 s^{2d-2}}, \tag{8}$$

$$1 - q_{M,s} \le 2^{2d+4}((d-1)!)^2 \frac{s^{2d-1}}{c_s^{2d-2}}. \tag{9}$$

If L is any generic contraction horizon satisfying $q_{M,s}^L \le \varepsilon < 1$, then

$$L \ge \frac{c_s^{2d-2}}{2^{2d+5}((d-1)!)^2 s^{2d-1}} \log(1/\varepsilon). \tag{10}$$

Proof. Let e_d be the last coordinate vector. The first coordinate of $A_s^j e_d$ is

$$\binom{j}{d-1} q_s^{j-d+1} c_s^{d-1}.$$

Put $n = \lceil s^{-1} \rceil$ and sum its square over $n \le j \le 2n$. On this range, $q_s^{2j} = (1-s)^j \ge 2^{-6}$, there are at least s^{-1} terms, and

$$\binom{j}{d-1} \ge \frac{(j/2)^{d-1}}{(d-1)!}.$$

This proves (7). Since $e_1^* M_s e_1 = \sum_j q_s^{2j} = s^{-1}$, one has $\lambda_{\min}(M_s) \le s^{-1}$, giving (8). Equation (9) follows from (3). Finally $q_{M,s} \ge 1/2$ and $-\log q \le 2(1-q)$ on $[1/2, 1)$; solving $q_{M,s}^L \le \varepsilon$ gives (10). □

Corollary 3.3 (Power-law ledger)**.** *Fix $d \ge 2$, $\kappa > 0$, and $0 \le \alpha < 1$. If $c_s = \kappa s^\alpha$, then as $s \downarrow 0$,*

$$\operatorname{cond}(M_s) \gtrsim s^{-2(d-1)(1-\alpha)}, \tag{11}$$

$$1 - q_{M,s} \lesssim s^{1+2(d-1)(1-\alpha)}, \tag{12}$$

$$L(\varepsilon) \gtrsim s^{-[1+2(d-1)(1-\alpha)]} \log(1/\varepsilon). \tag{13}$$

At the threshold $\alpha = 1$, Theorem 3.1 gives bounded conditioning but still a generic $s^{-1}\log(1/\varepsilon)$ horizon.

Corollary 3.4 (Growing-chain barrier)**.** *Suppose $c_s \ge c_0 > 0$ and $d_s \ge a \log(1/s)$ while $d_s = O(\log(1/s))$. Then*

$$\log \operatorname{cond}(M_s) \ge (2a + o(1))(\log(1/s))^2. \tag{14}$$

In particular, the full metric condition number is super-polynomial in s^{-1}.

Proof. Use $((d-1)!)^{1/(d-1)} \le d-1$ in (8), take logarithms, and substitute the assumed growth of d_s. The $\log d_s$ term is lower order than $\log(1/s)$. □

4 An exact two-step witness

For $d = 2$, direct solution of (2) gives

$$M_s = \begin{pmatrix} s^{-1} & \dfrac{q_s c_s}{s^2} \\ \dfrac{q_s c_s}{s^2} & s^{-1} + c_s^2(1+q_s^2)s^{-3} \end{pmatrix}. \tag{15}$$

Its determinant is

$$\det M_s = s^{-2} + c_s^2 s^{-4}. \tag{16}$$

For fixed $c_s = c > 0$,

$$\lambda_{\max}(M_s) \sim 2c^2 s^{-3}, \quad \lambda_{\min}(M_s) \sim (2s)^{-1}, \quad \operatorname{cond}(M_s) \sim 4c^2 s^{-2}, \quad 1 - q_{M,s} \sim \frac{s^3}{4c^2}. \tag{17}$$

This two-dimensional example already rules out the idea that a global Lyapunov change of norm can erase a nonnormal near-peripheral time scale.

5 High-precision family audit

We solve the vectorized Lyapunov equation with 140 decimal digits at $s = 10^{-2}, \dots, 10^{-8}$. The coupling law is $c_s = 0.2s^{\alpha}$. Exponents are least-squares slopes over the final four points. The predicted columns are the powers in Corollary 3.3; the agreement is diagnostic, not part of the proof.

Table 1: Predicted and fitted powers as $s \downarrow 0$.

d	α	cond pred.	cond fit	gap pred.	gap fit
2	0.00	2.0	2.000001	3.0	3.000001
3	0.00	4.0	4.000003	5.0	5.000001
4	0.00	6.0	6.000004	7.0	7.000002
4	0.50	3.0	3.000027	4.0	3.999984
4	0.75	1.5	1.505607	2.5	2.494042
4	1.00	0.0	0.000000	1.0	1.000000

At $d = 4$, $s = 10^{-8}$, fixed coupling has $\mathrm{cond}(M_s) = 5.12\times 10^{45}$ and $1-q_{M,s} = 3.90625\times 10^{-54}$. Gap-scale coupling has condition number 1.94698 and $1 - q_{M,s} = 3.39584 \times 10^{-9}$, but even this favorable row needs approximately 4.07×10^9 generic contraction steps to reach 10^{-6}. Thus conditioning control is necessary but not sufficient for a polylogarithmic horizon.

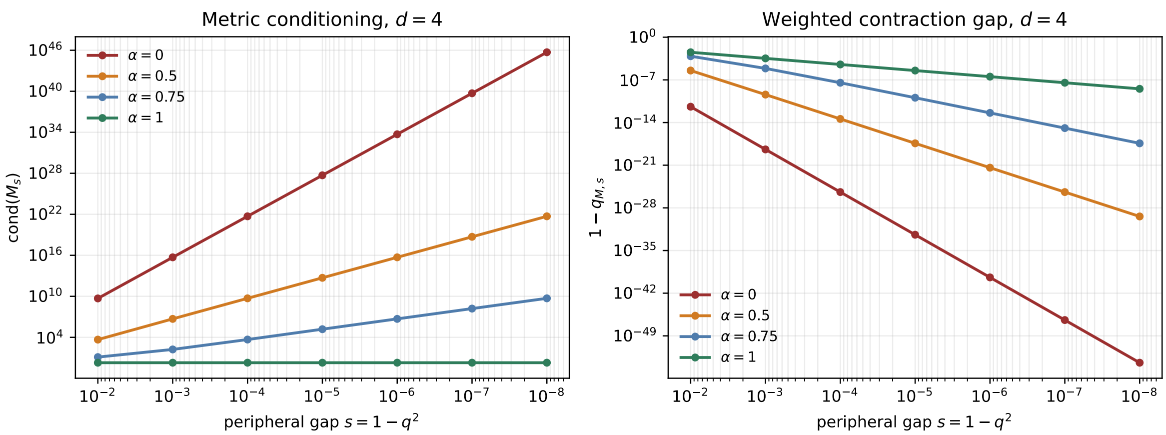

Figure 1: Dimension-four coupling laws $c_s = 0.2s^{\alpha}$. Below the gap-scale threshold, conditioning grows and the weighted contraction gap loses exactly one additional power for every conditioning power.

At $s = 10^{-4}$, the 256-bit Arb audit certifies the exact two-step Lyapunov identities and positivity. For fixed $c = 0.2$, it certifies $\mathrm{cond}(M_s) > 10^7$ and $1 - q_{M,s} < 10^{-10}$. For matched $c = 0.2s$, it certifies $\mathrm{cond}(M_s) < 2$ and $0.4 < (1 - q_{M,s})/s < 0.5$. These are interval-certified family points, not a production interval theorem.

6 Route consequence

The result closes one ambiguity in RH-64. Recomputing a canonical metric on the full near-peripheral space is not a uniform cure: above the coupling-gap threshold it introduces polynomial or super-polynomial costs, and at the threshold it still sees the original slow time scale. This is a negative result for the order

$$\text{full physical space} \longrightarrow \text{one global metric} \longrightarrow \text{tail}.$$

It does not obstruct three narrower constructions:

1. use Krylov projection to remove the reached slow directions before building the terminal metric;
2. use block metrics whose coupling is measured relative to each local spectral gap;
3. use observation-weighted Gramians that ignore invisible global directions.

The next gate is therefore a block, cross-column Krylov Gram certificate. It must retain the coherent phase fusion of RH-58–RH-60 while applying RH-64 only to the final localized residual blocks.

7 No arithmetic or Hilbert–Polya conclusion

This paper constructs no self-adjoint operator, no $T \log T$ counting law, no prime-power trace formula, and no completed-zeta identity. It makes no Hilbert–Polya or Riemann-hypothesis claim. Stage A1 and unconditional Stage A4 remain open.

8 Reproducibility

The directory contains the high-precision solver, tests, family pilot, Arb audit, figures, hashes, and publication artifacts. The principal commands are:

```
pytest -q -p no:cacheprovider
python experiments/run_family_conditioning_pilot.py
python experiments/run_arb_two_step_audit.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The matrix identities and bounds are analytic. The fitted exponents are high-precision finite-family evidence. Uniform folded-Gaussian metric conditioning, block cross-column fusion, Stage A1, unconditional Stage A4, a self-adjoint Hilbert–Polya operator, an arithmetic trace formula, and a zeta-zero identity remain open.

References

[1] Roger A. Horn and Charles R. Johnson. *Matrix Analysis.* Cambridge University Press, 2 edition, 2013.

[2] Lloyd N. Trefethen and Mark Embree. *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators.* Princeton University Press, Princeton, 2005.

[3] Liang Wang. Observability-weighted terminal residuals for nested Krylov Stein tails. Preprint and software repository, 2026. URL `https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-64-weighted-terminal-residuals`.

[4] Kemin Zhou, John C. Doyle, and Keith Glover. *Robust and Optimal Control.* Prentice Hall, Upper Saddle River, NJ, 1996.